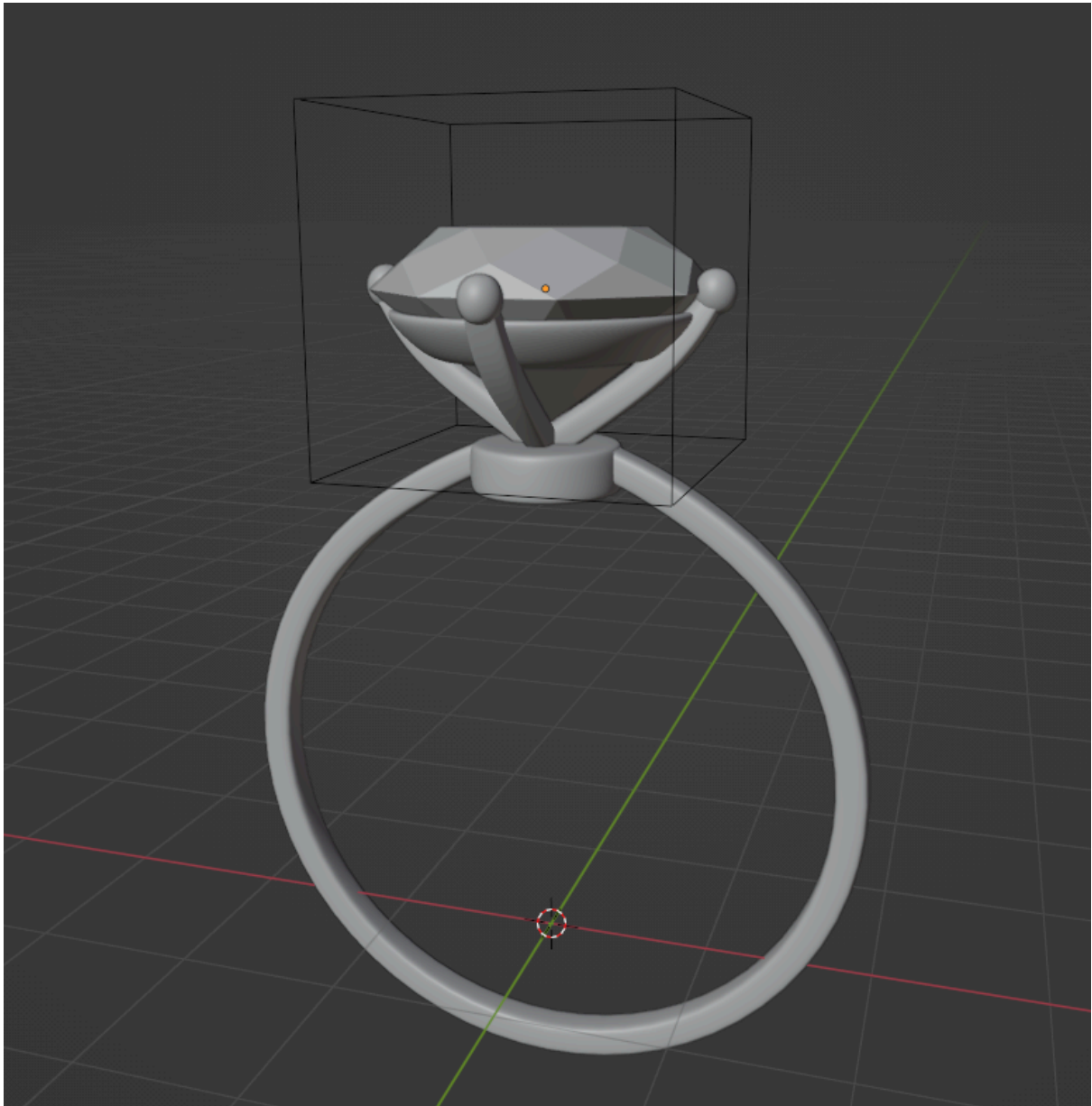
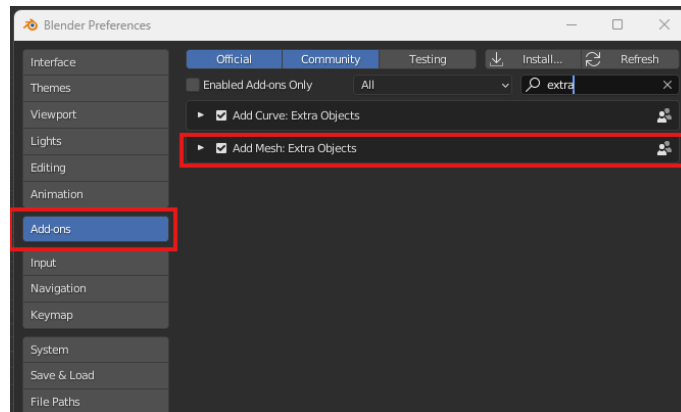


Ring tutorial

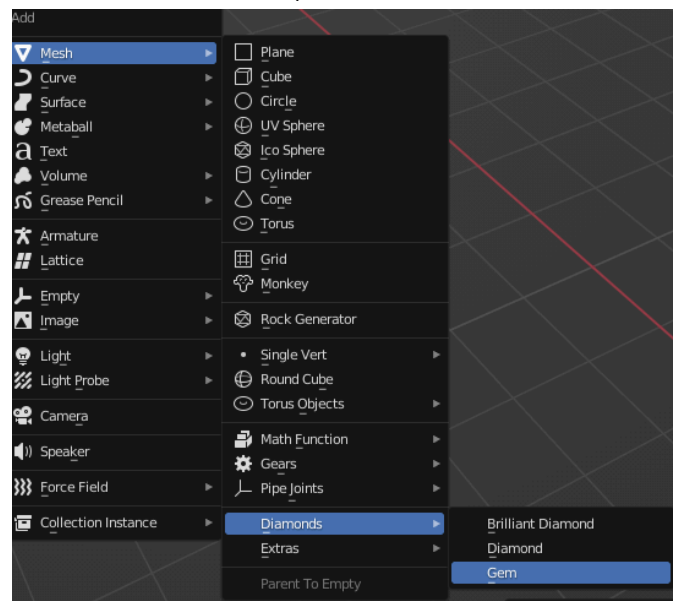
Important concepts covered: add mesh, knife tool, extrude, add loop cut.



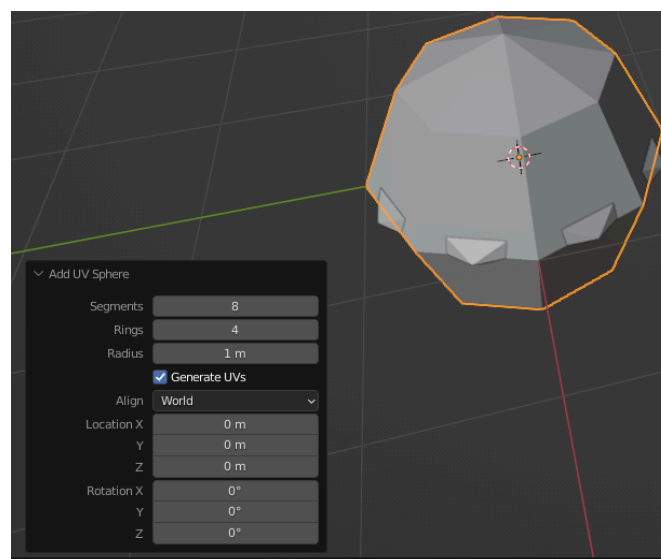
First, go to Edit > Preferences > Add-ons and enable “Add Mesh: Extra Objects”



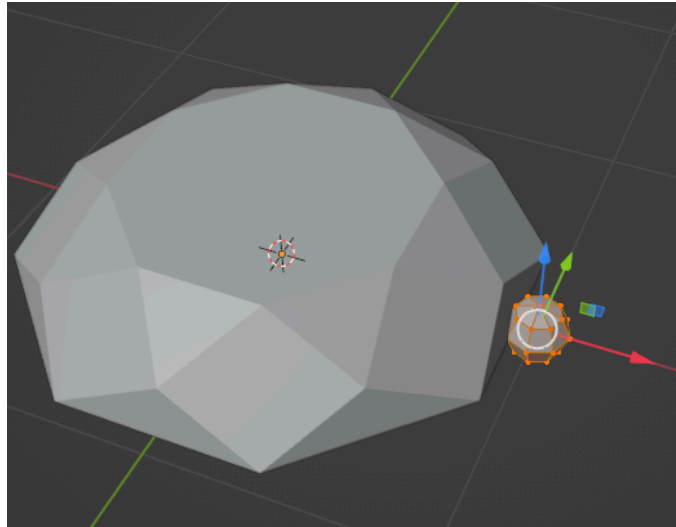
Before using the recommender system, save the blender project in the computer. Select all and delete and then add a “Gem” under Diamonds category under add mesh (SHIFT+A → MESH → Diamonds → Gem):



Add a UV Sphere (SHIFT+A → MESH → UV Sphere) and set the Segments to 8 and Rings to 4:



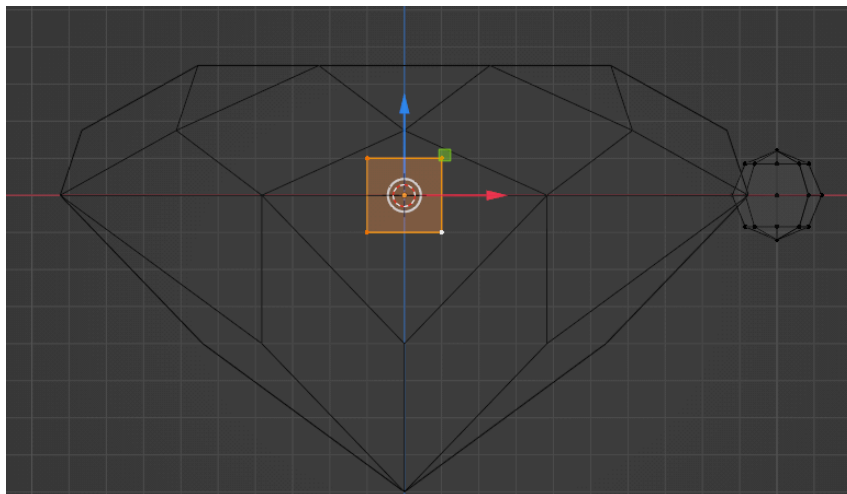
Enter edit mode on the UV Sphere and scale it (S) down by 0.12 and move along X axis (G+X) by 1:



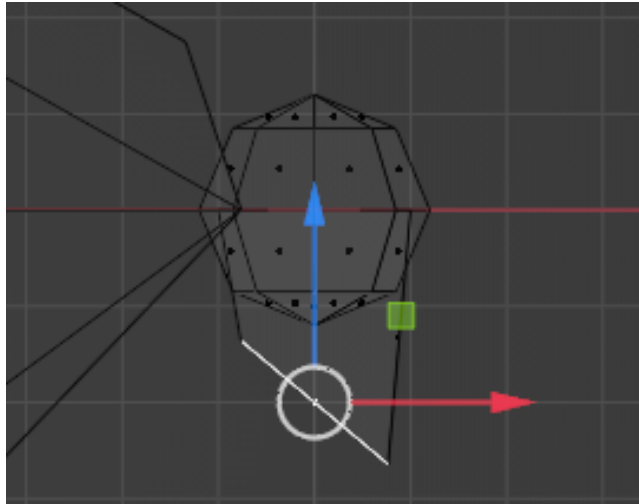
Go to the front view (NUMPAD 1) and enter wireframe mode by pressing “Z” and selecting “Wireframe”.

Next, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it).

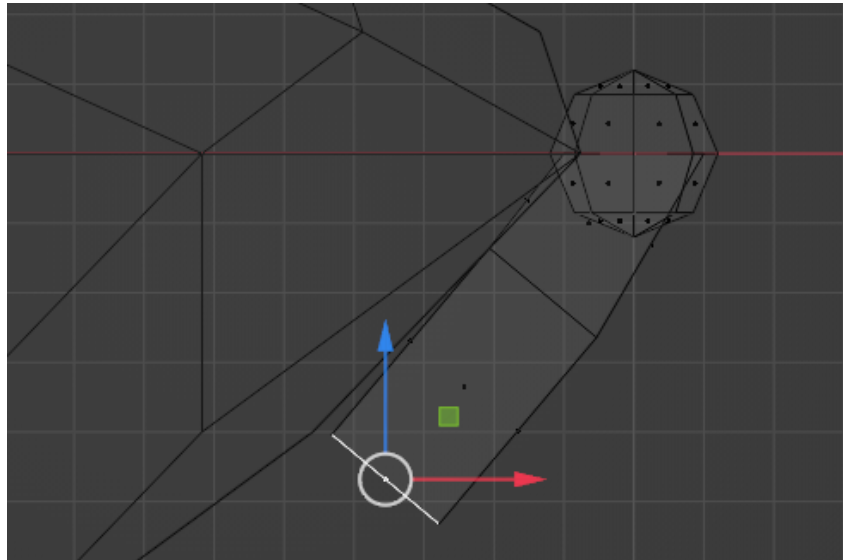
Then Add a Cube (SHIFT+A → MESH → Cube) and scale it (S) down by 0.1:



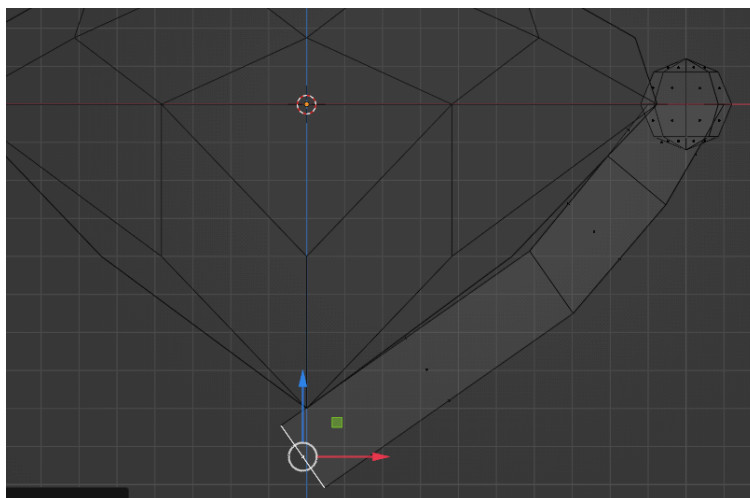
Move along X axis (G+X) by 1 and Z axis (G+Z) by -0.1, so the cube is under the gem, and change to Face selection mode. Then, select the bottom face of the cube and rotate (R) it by 40 degrees:



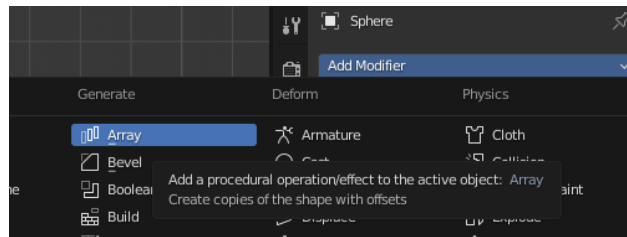
Move it along the X axis (G+X) by -0.13 so it touches the Diamond. Now extrude (E) 0.35 in the normal direction:



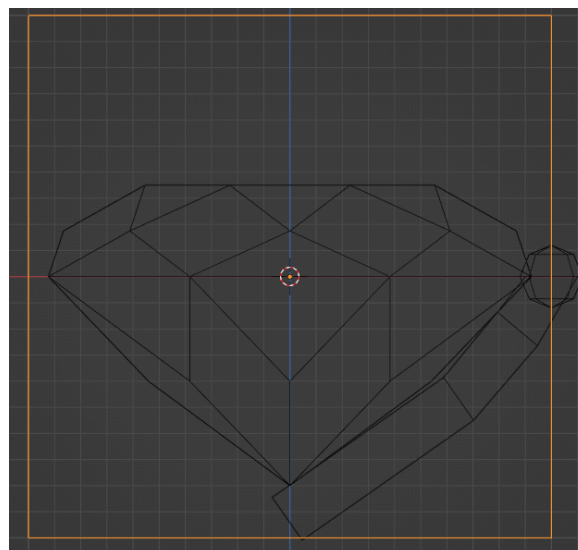
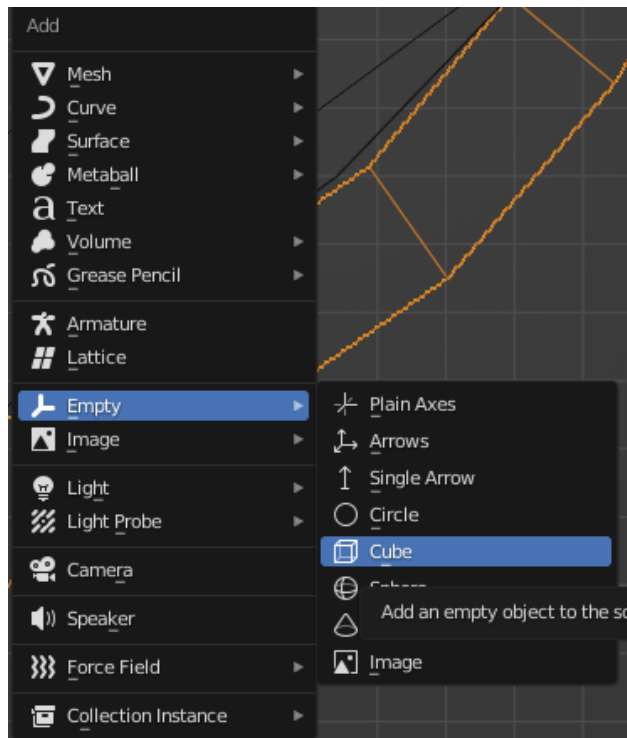
Now rotate (R) again by 15 degrees and extrude (E) 0.8 along the normal:



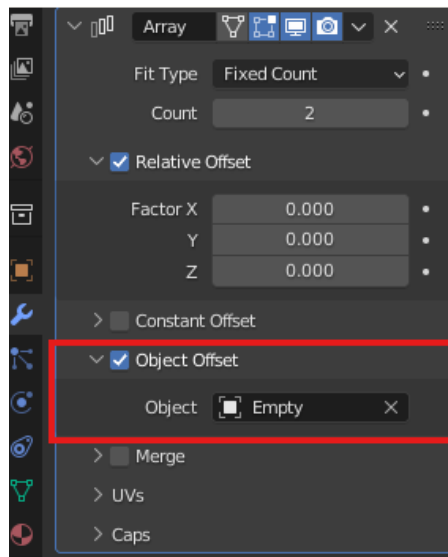
Now change back to object mode and add an array modifier to the created support of the gem:



Change the “Factor X” property to 0. We want to copy this support and spin it so it goes around the Diamond. To do this with the array modifier, we first need an empty object to act as an origin of the spin. So, next, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a new empty Cube (SHIFT+A → EMPTY → Cube):

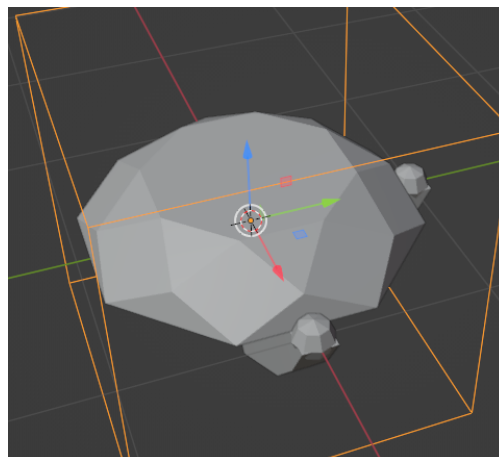


Select back the support and check “Object Offset” under the properties of the array modifier and pick the empty cube created:



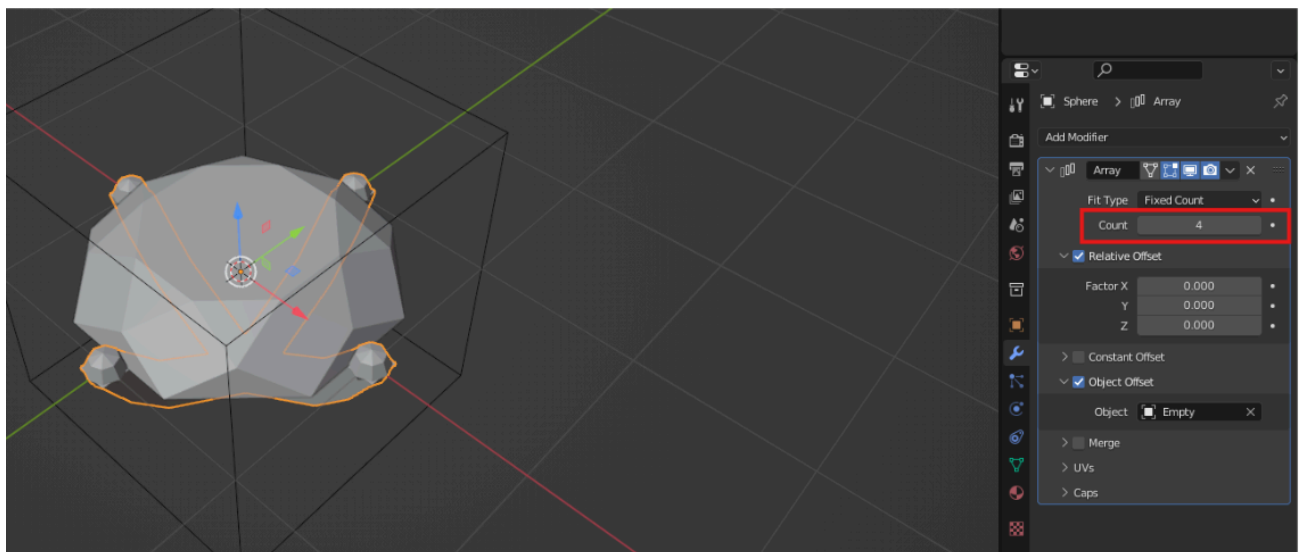
Change back to solid shade view by pressing “Z” and changing to “Solid”.

Now, the support will “follow” the offset of the empty cube. So to create the spin and repeat the support every 90 degrees, we can select the empty cube and rotate it along the Z axis (R+Z) by 90 degrees:

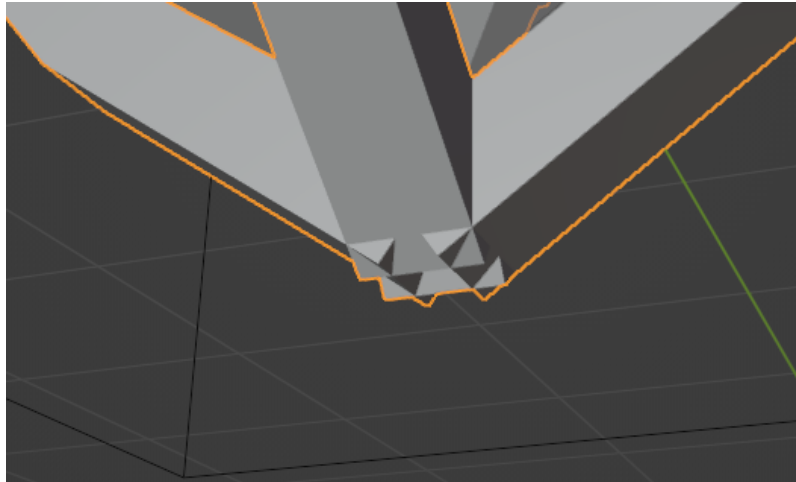


As you can see, a copy of the support was created with an offset of 90 degrees.

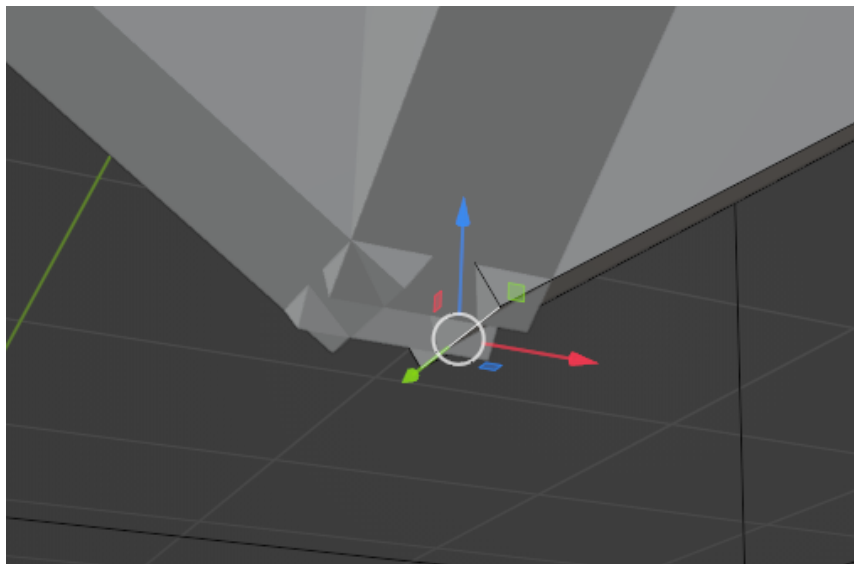
Now we can increase the count to 4 in the array modifier of the support:



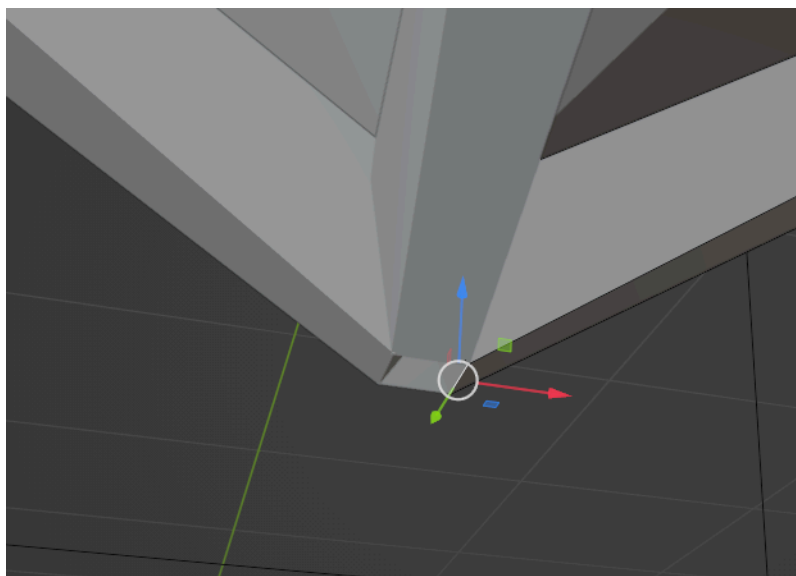
If you look under the Diamond, there is a problem in the mesh:



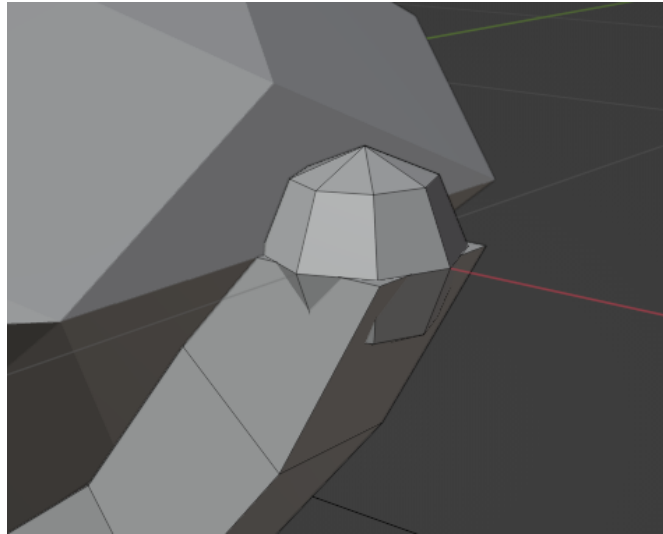
To correct it, we can enter edit mode again, change to Edge selection mode and select the edge creating the problem:



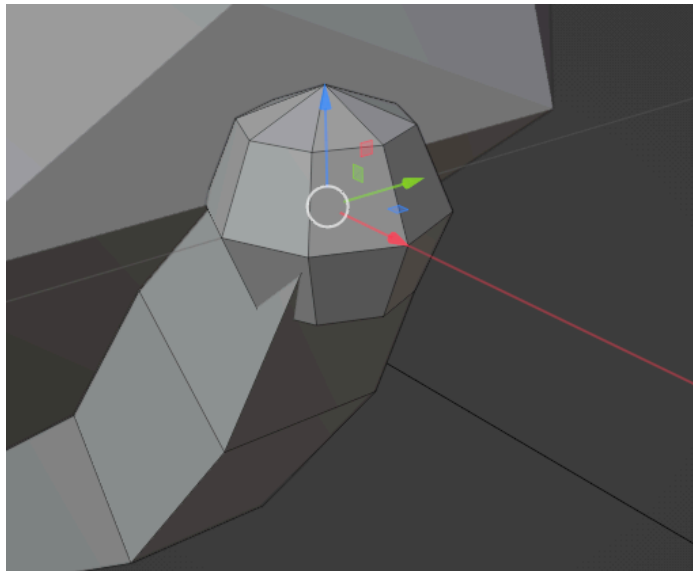
With this edge selected, we can scale it (S) down by 0.5 so it does not overlap with the copies:



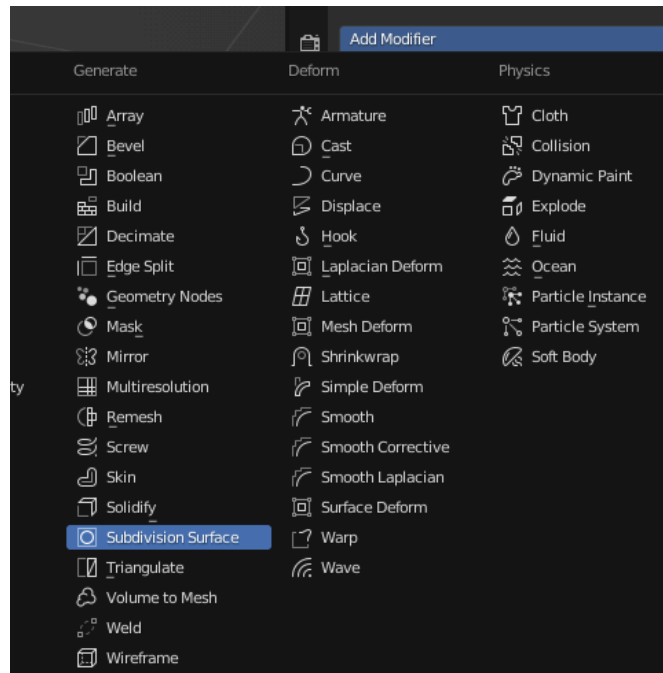
Another problem in the mesh is that the small gem is passing through the support:



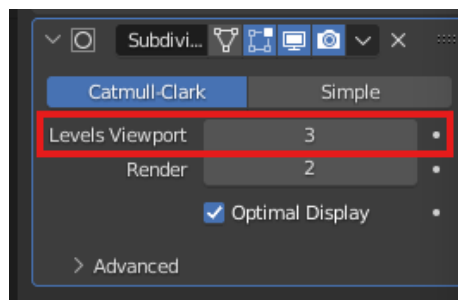
To correct it, select the top face of the support and scale (S) it down by 0.8:



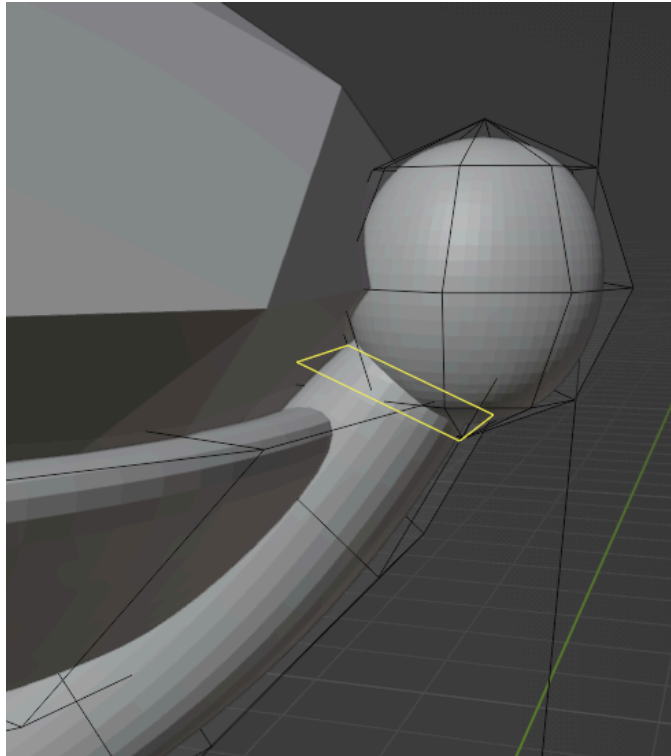
Now we can add a Subdivision Surface modifier to see how it looks with more polygons:



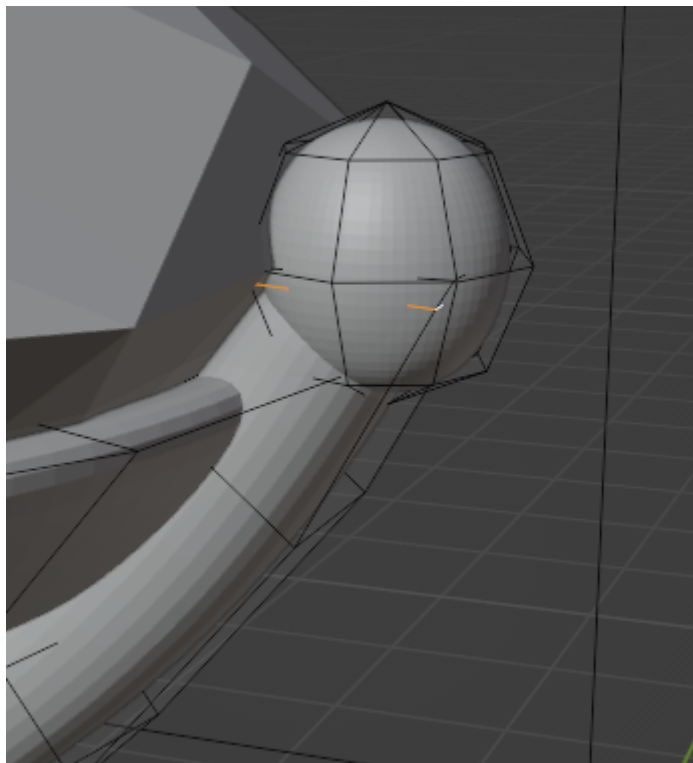
Set the “Levels Viewport” to 3:



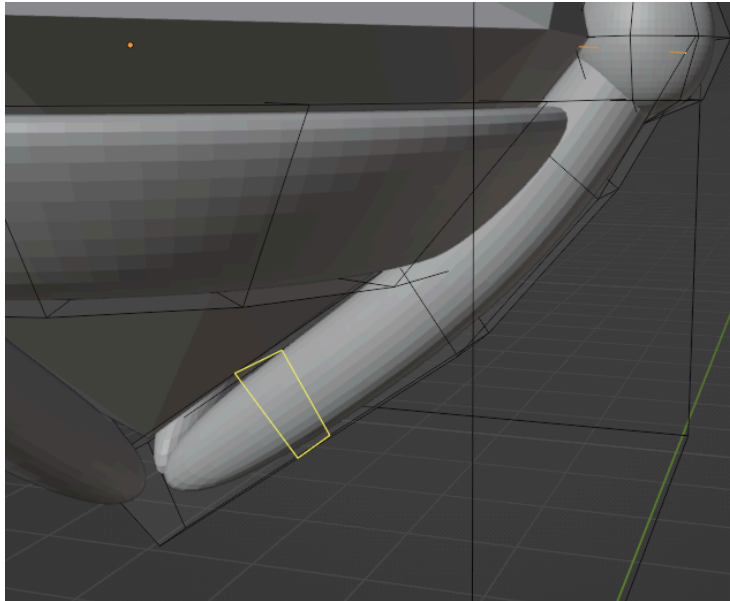
Now we can start working on its final look. First, let's add a Loop cut on the top part of the support and move the cut closer to its end. To do this, press CTRL+R, move the mouse pointer to the center of the upper part of the support and left click once:



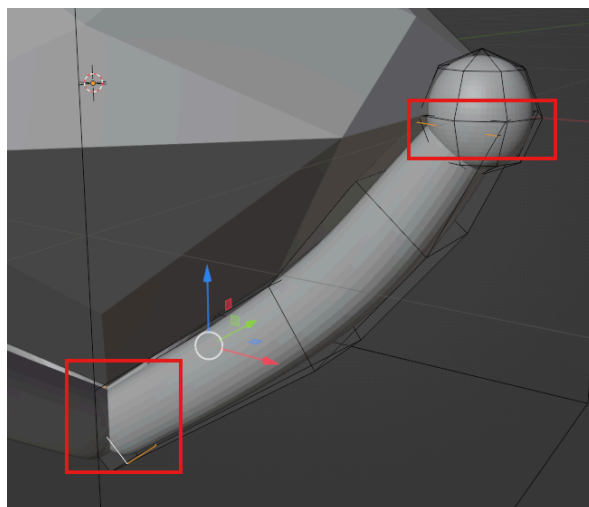
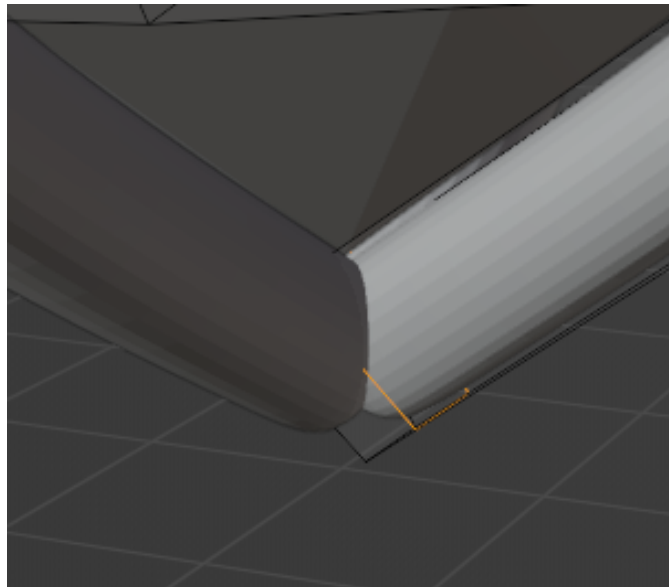
Then, type -0.8 and left click again to confirm:



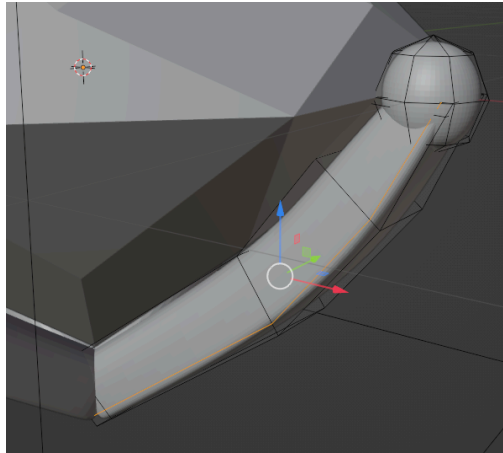
Now, let's add a Loop cut on the bottom part of the support and move the cut closer to its end, like we did to the upper part. So, again, press CTRL+R, move the mouse pointer to the center of the bottom part of the support and left click once:



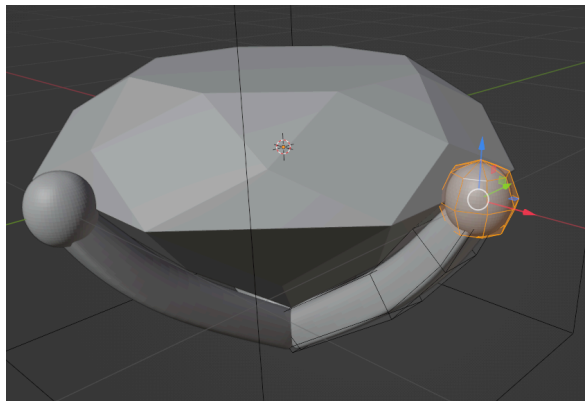
Then, type -0.8 and left click again to confirm:



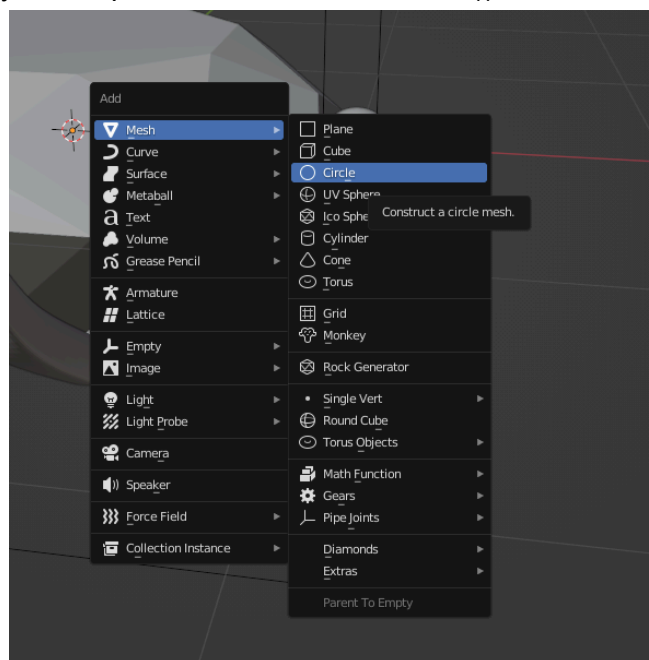
Now, let's add another Loop Cut (CTRL+R) but now along the side face of the support. Set the Factor as -0.7:

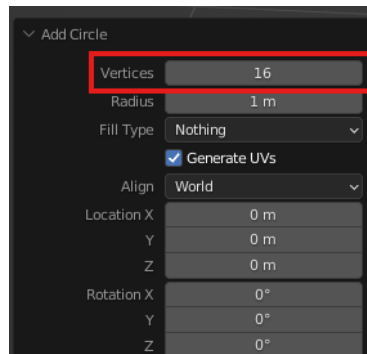


Since the gem looks too small, select it (you can do this by De-selecting everything by double tapping “A” and then pointing the mouse towards the UV Sphere of the gem and pressing “L” to select the linked mesh) and scale it (S) up by 1.2:



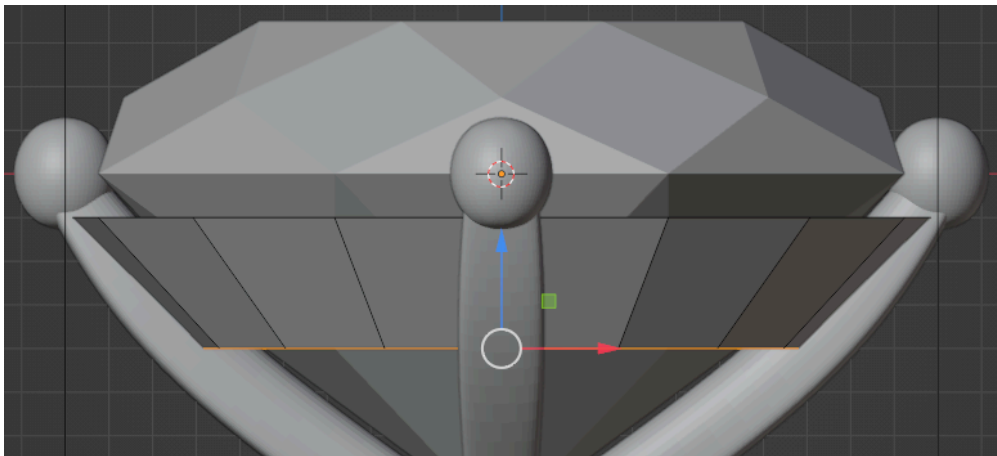
Now lets create another detail around the Diamond. Go back to Object mode and create a new mesh Circle with the number of vertices set to 16 (make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it)):



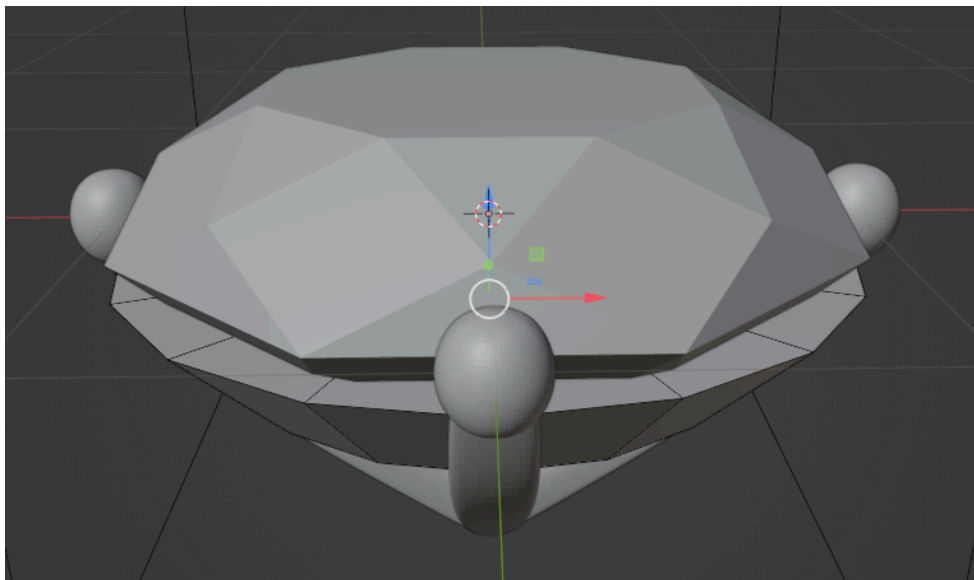


Now go to the front view and enter edit mode on the created circle object. Move the circle along the Z axis (G+Z) by -0.1.

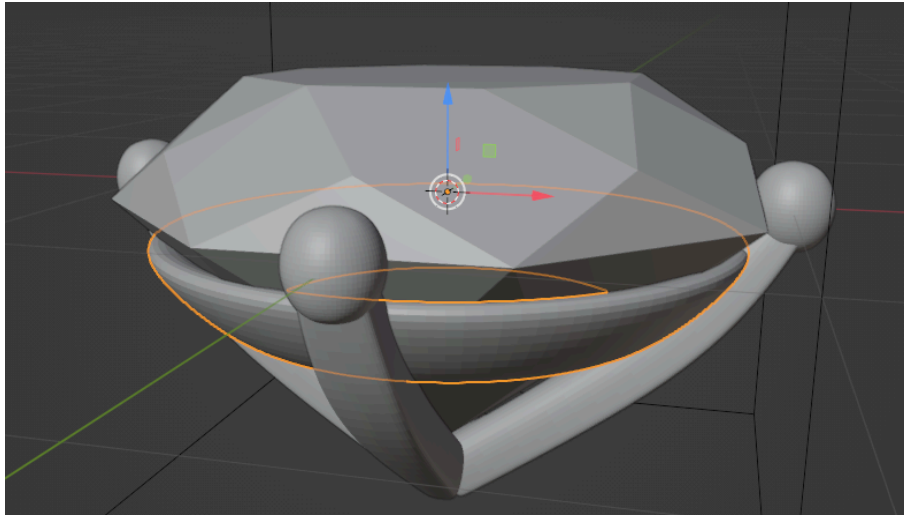
Now extrude it and move along the Z axis (G+Z) -0.3. Then scale it (S) down by 0.7:



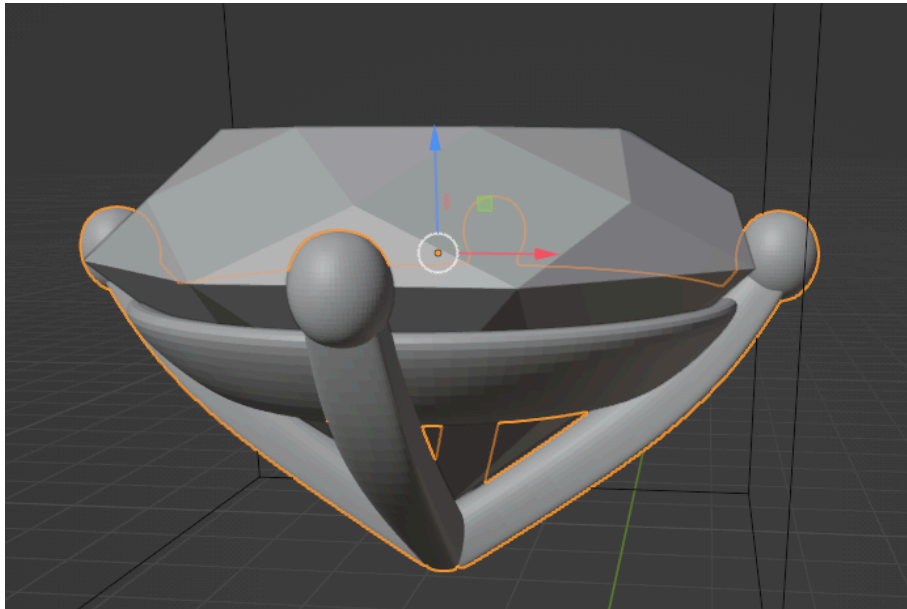
Now select all, extrude (press E) and then scale down in all directions except Z (you can do this by pressing "S" to scale and then SHIFT+Z to ignore Z axis) by 0.8:



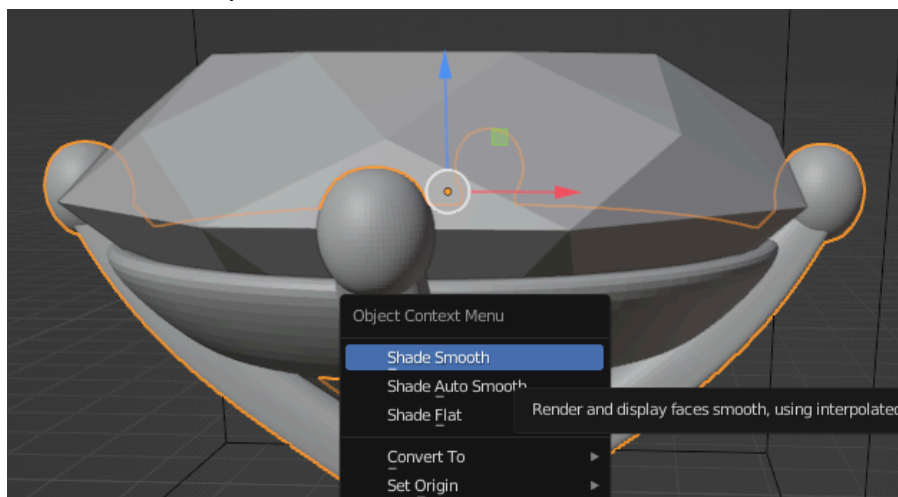
Now go to object mode and add a Subdivision Surface modifier and set to 3 the “Levels Viewport” property:



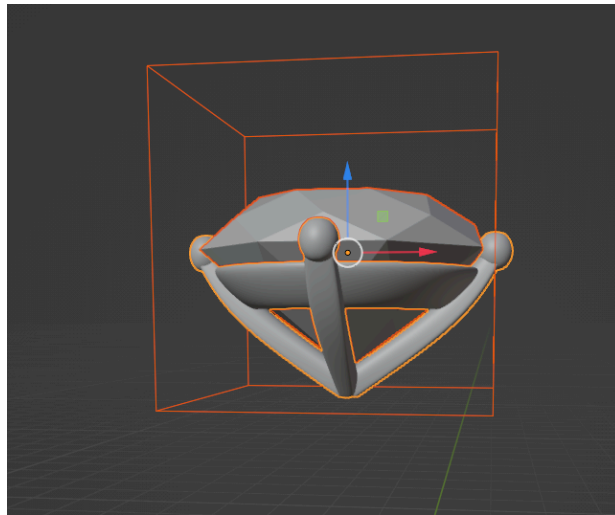
Next, with this created circle selected, select also the support and hit “CTRL+J” to merge these objects together:



Finally, right click on the viewport and select “Shade Smooth”:

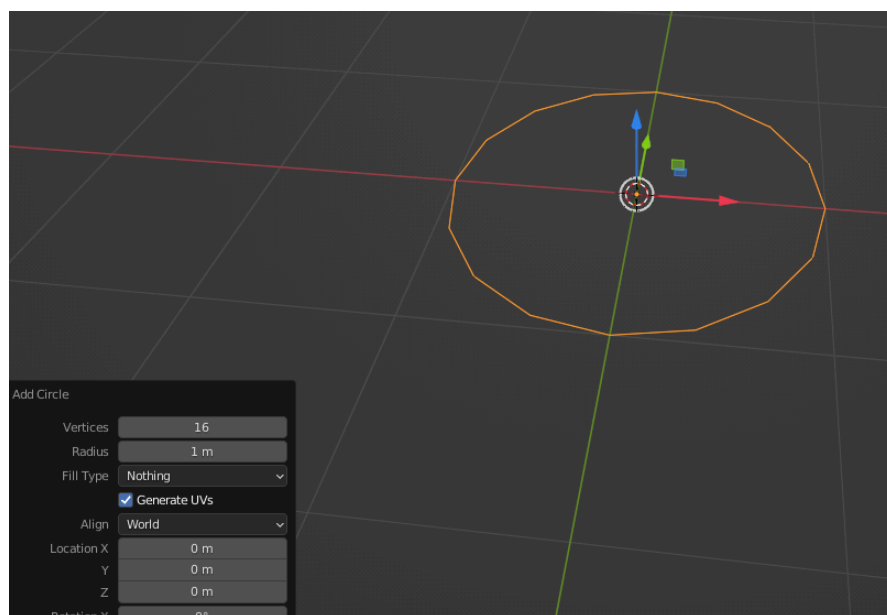


Now, select all in the scene by pressing “A” and move it along the Z axis (G+Z) by 4:

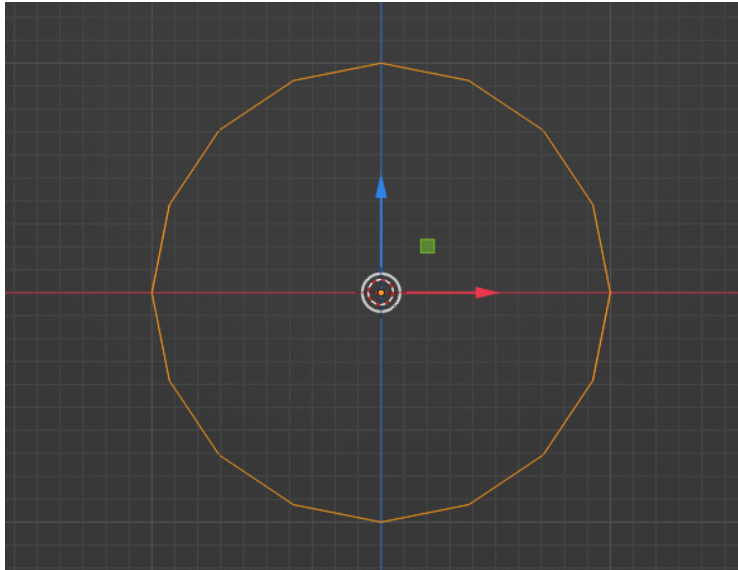


Perfect! Now we have the top part of the ring completed. Let's start now modeling the bottom part of the ring (the part where the finger is put).

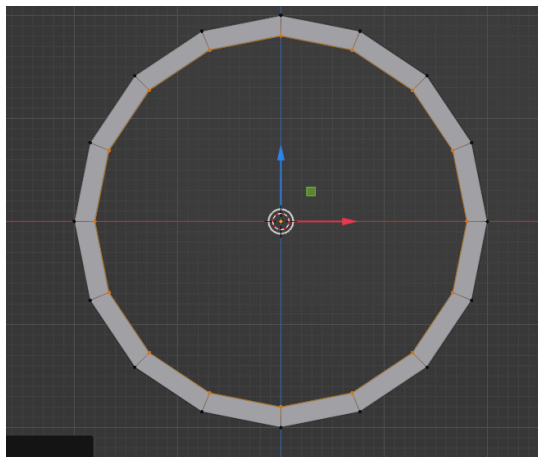
First, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a new mesh Circle and make sure that the number of Vertices is set to 16:



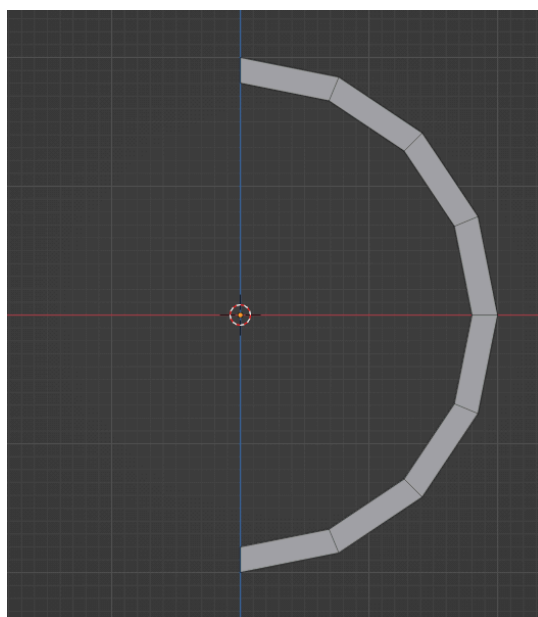
Go to the front view (NUMPAD 1), enter edit mode and rotate in the X axis (R+X) by 90 degrees:



Now scale it (S) up by 2 and extrude (E) it and scale it (S) down by 0.9:

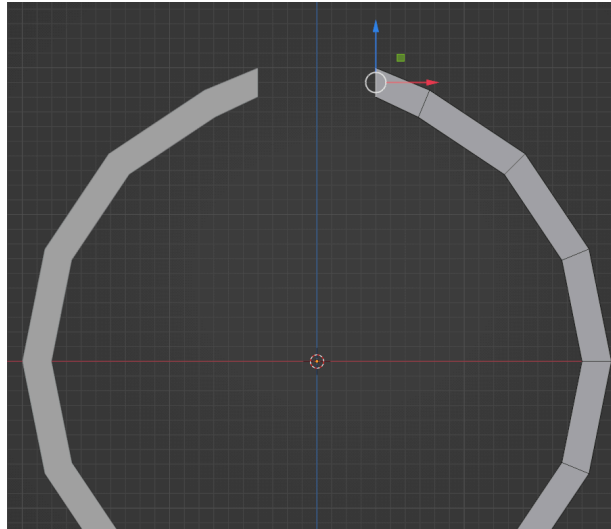


Now, change to face selection mode and select all the left part of the bottom ring and delete it:

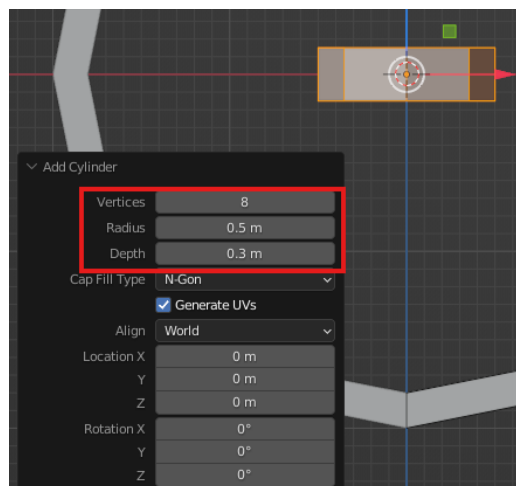


Then add a mirror modifier. The bottom ring should now be “complete” again.

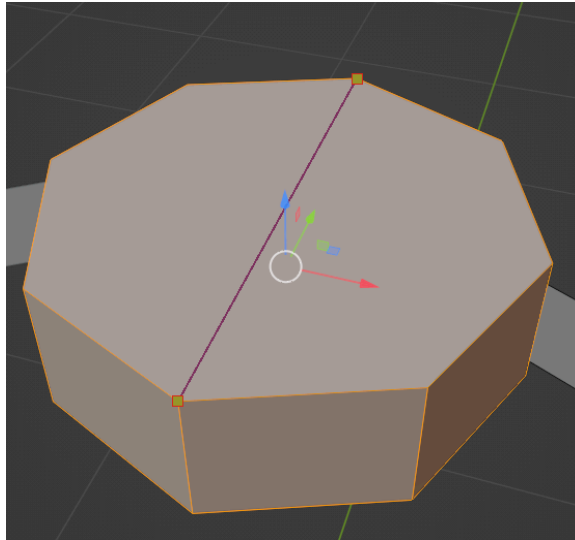
Next, change to edge selection mode again and select the top vertical edge and move it along the X axis (G+X) by 0.4, so the bottom ring is now “Open”:



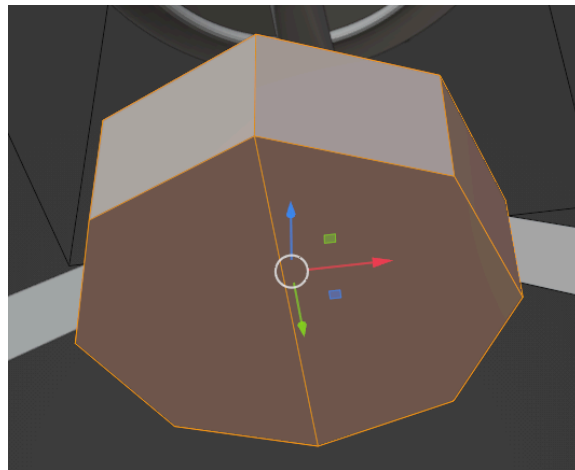
Now, make sure that the 3D cursor is at the center of the scene (you can press SHIFT + C to center it) and add a cylinder (SHIFT+A → MESH → Cylinder) and set its vertices to 8, Radius to 0.5 and Depth to 0.3:



Move it along the Z axis (G+Z) by 1.86 so it is right in the middle of the created opening. Since we have the mirror modifier, we have to delete the left side of the cylinder. In order to do so, you can select all the left faces and delete them. However, the top faces cannot be deleted on only one side, since they are round. So first, cut the top and bottom faces in the middle by using the knife tool (shortcut “K”). Select first vertex and then the second one and then hit “Enter” to confirm the cut:

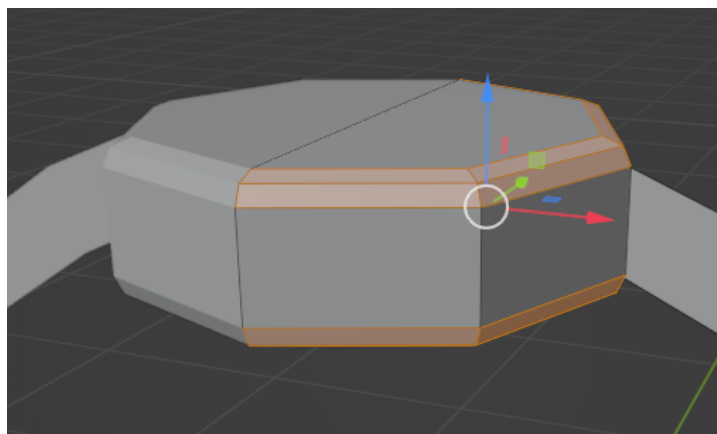


Do this also to the bottom face:

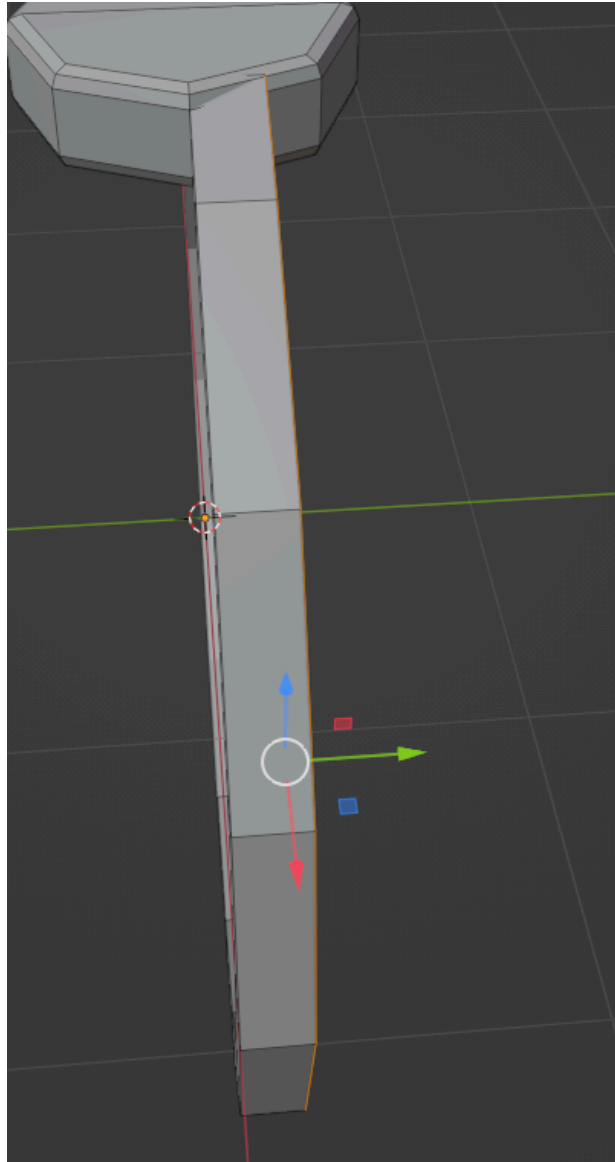


Now you can select all the left faces and delete them. As you can see, the deleted part is still there because it is mirrored. Now let's add some details to this cylinder.

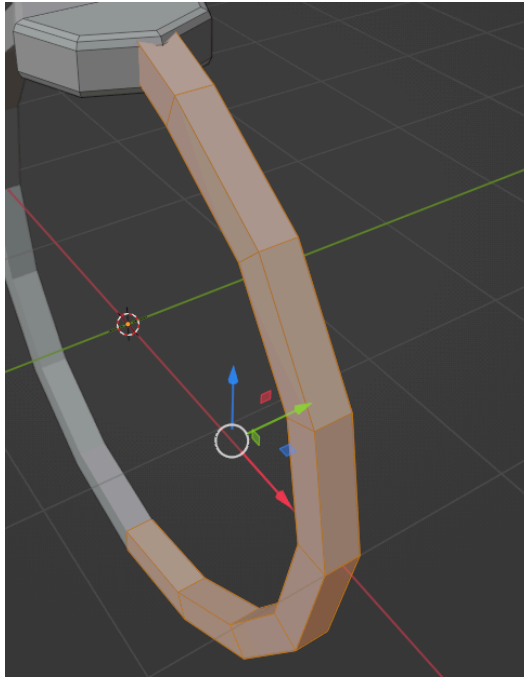
First, select the top and bottom faces and bevel them (Shortcut "CTRL+B") by 0.05 and 2 segments:



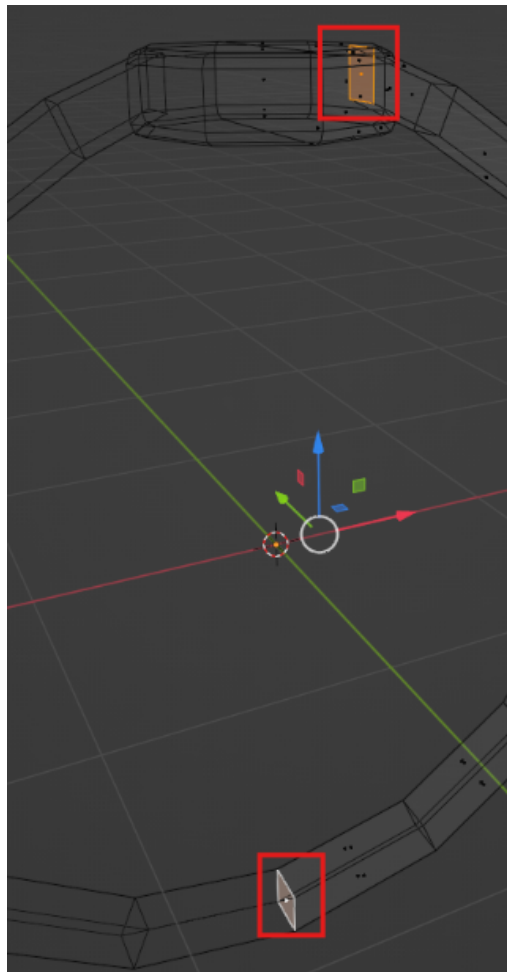
Now, let's add some depth to the bottom ring. Select it (you can double tap "A" to de-select all and use "L" again to select linked mesh pointing to the bottom ring) and extrude it in the Y axis (E+Y) by 0.2:



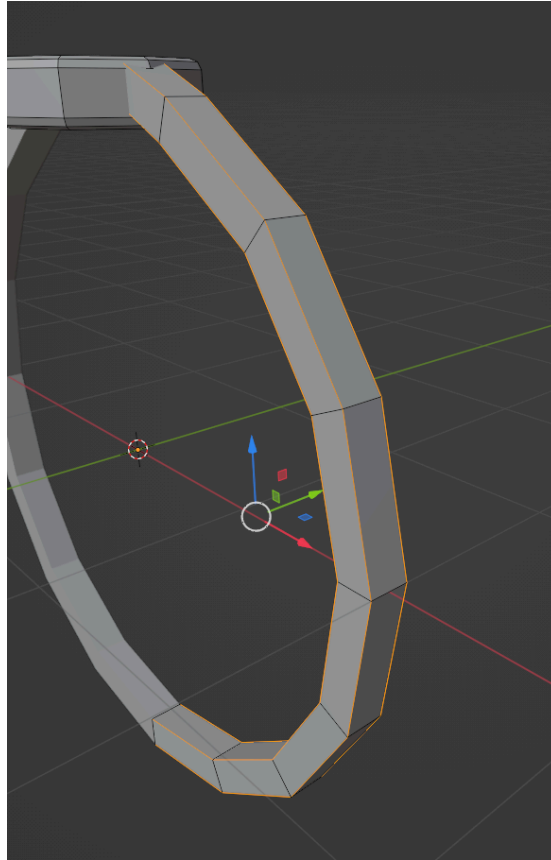
Now select all the bottom ring again and move it in the Y axis (G+Y) by -0.1 to center it again:



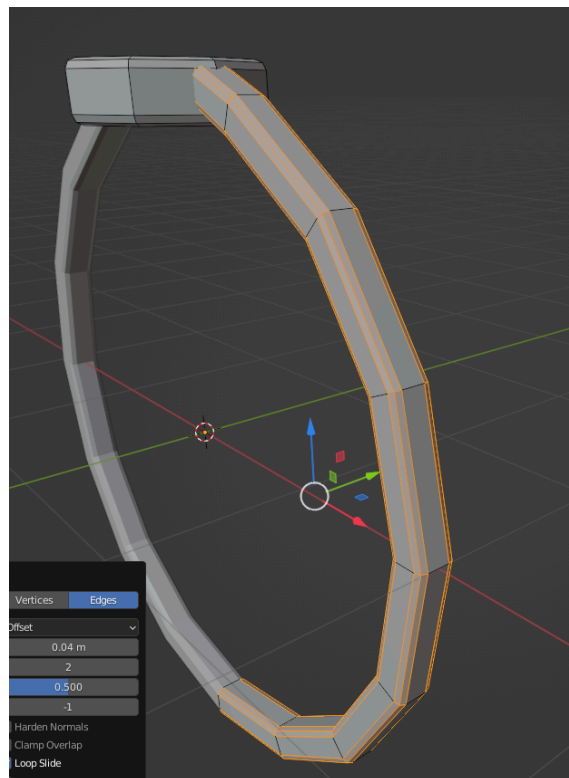
Now go to face mode selection, change shading view to wireframe, select the inside faces of the bottom ring and delete (DEL or X) them:



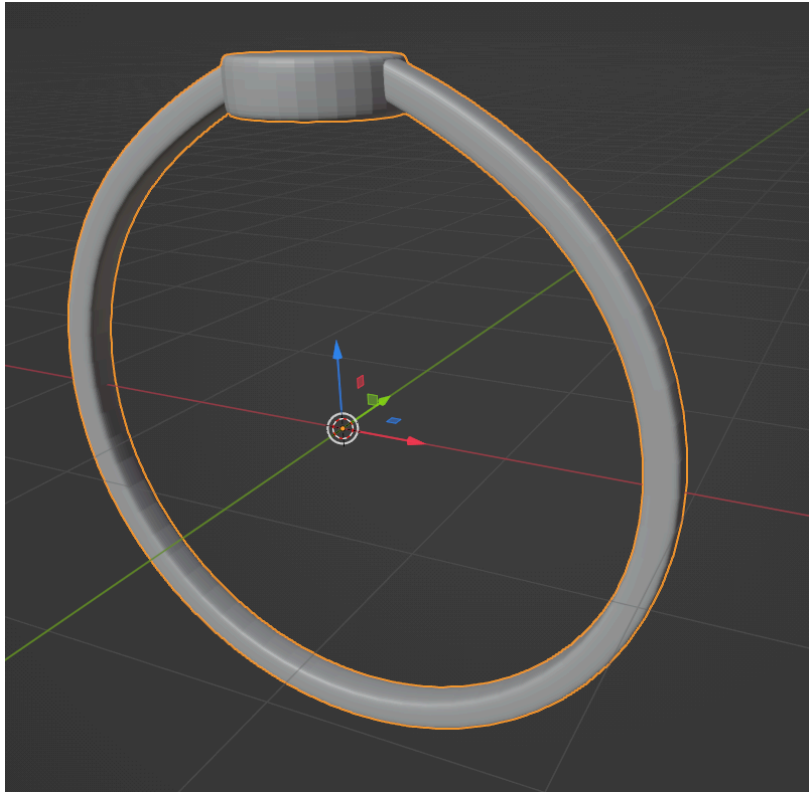
Change back to Solid view and select all the outer edges of the bottom ring. You can do this by holding “ALT” and then left clicking on one edge to select the loop. Then, to select the next loops, hold “ALT+SHIFT” and continue selecting the edges:



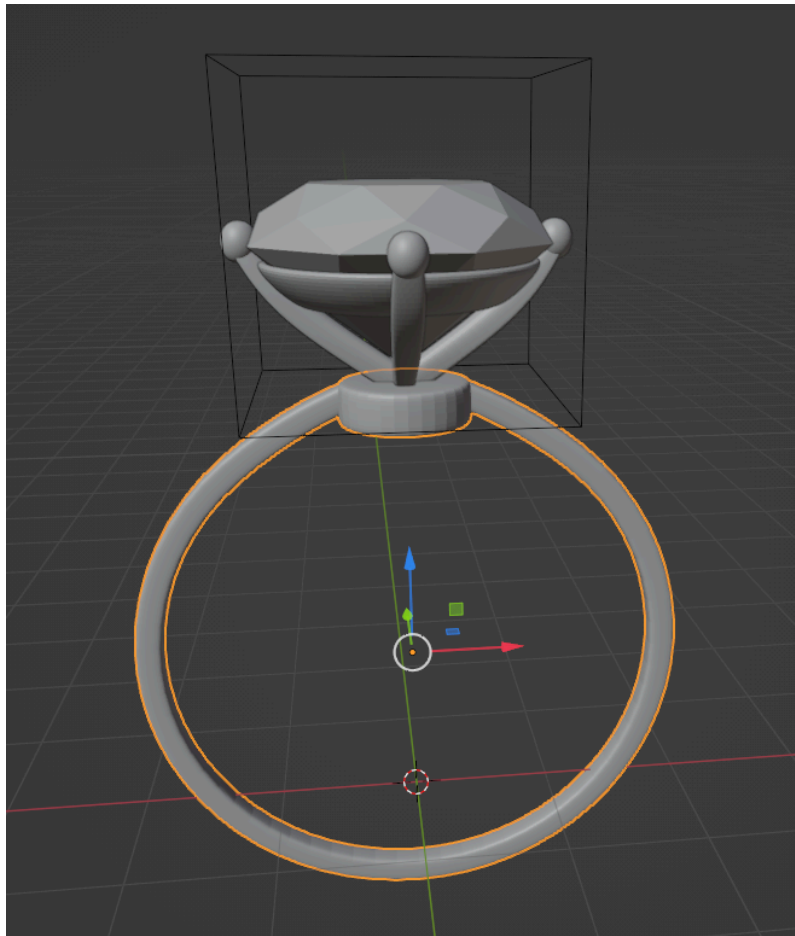
Now bevel it (CTRL+B) by 0.04 with 2 segments as before:



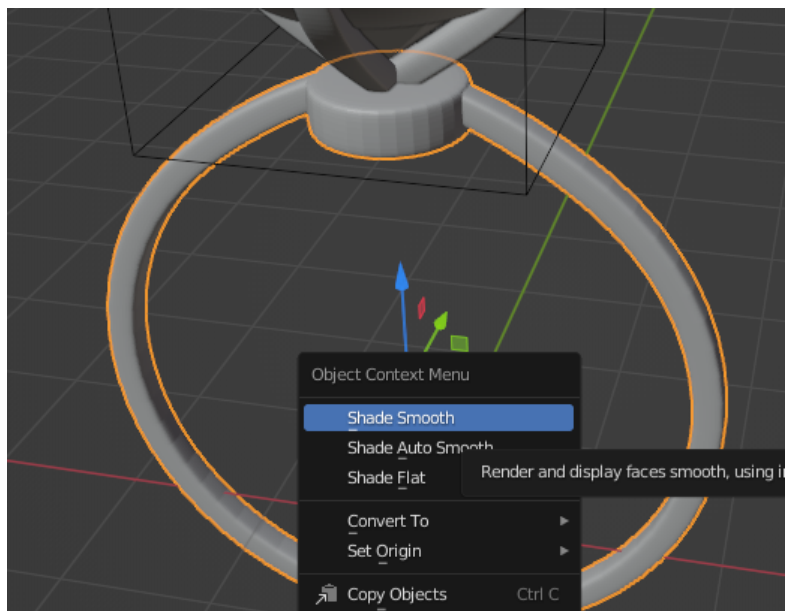
Next, go to object mode and then add a Subdivision Surface modifier setting the levels to 2:



Now to conclude the whole ring (bottom + upper part), move the bottom ring along Z axis (G+Z) by 1.1:



Finally, with the bottom ring selected, right click on the viewport and select “Shade Smooth”:



Now you have the whole ring completed!:

